



Altman Z-Score Presentation

This presentation will provide an introduction to the Altman Z-Score financial analysis model.

	B	C	D
		Company's View	
	250	Public Manuf Z-Sco	
	20	SECCOMP	
	80		
	160	Private Manuf Z-Sco	
	70	SECCOMP	
	120		
	20	General Z-Sco	
		SECCOMP	
y	250		
	150		
THRESHOLDS	Public Manuf	Private Man	
y sound if >=	0		
Caution			
inruptcy if <	0		

What is the Altman Z-Score?

The Altman Z-Score is a formula developed in 1968 by Edward Altman to predict the likelihood that a company will go bankrupt within two years. It analyzes several financial ratios to determine the overall financial health of a company.

History

● **1966**

Edward Altman joined New York University as an Assistant Professor of Finance.

● **1967**

Altman began researching bankruptcy prediction models.

● **1968**

Altman published his Z-Score formula in the Journal of Finance.

● **1970s**

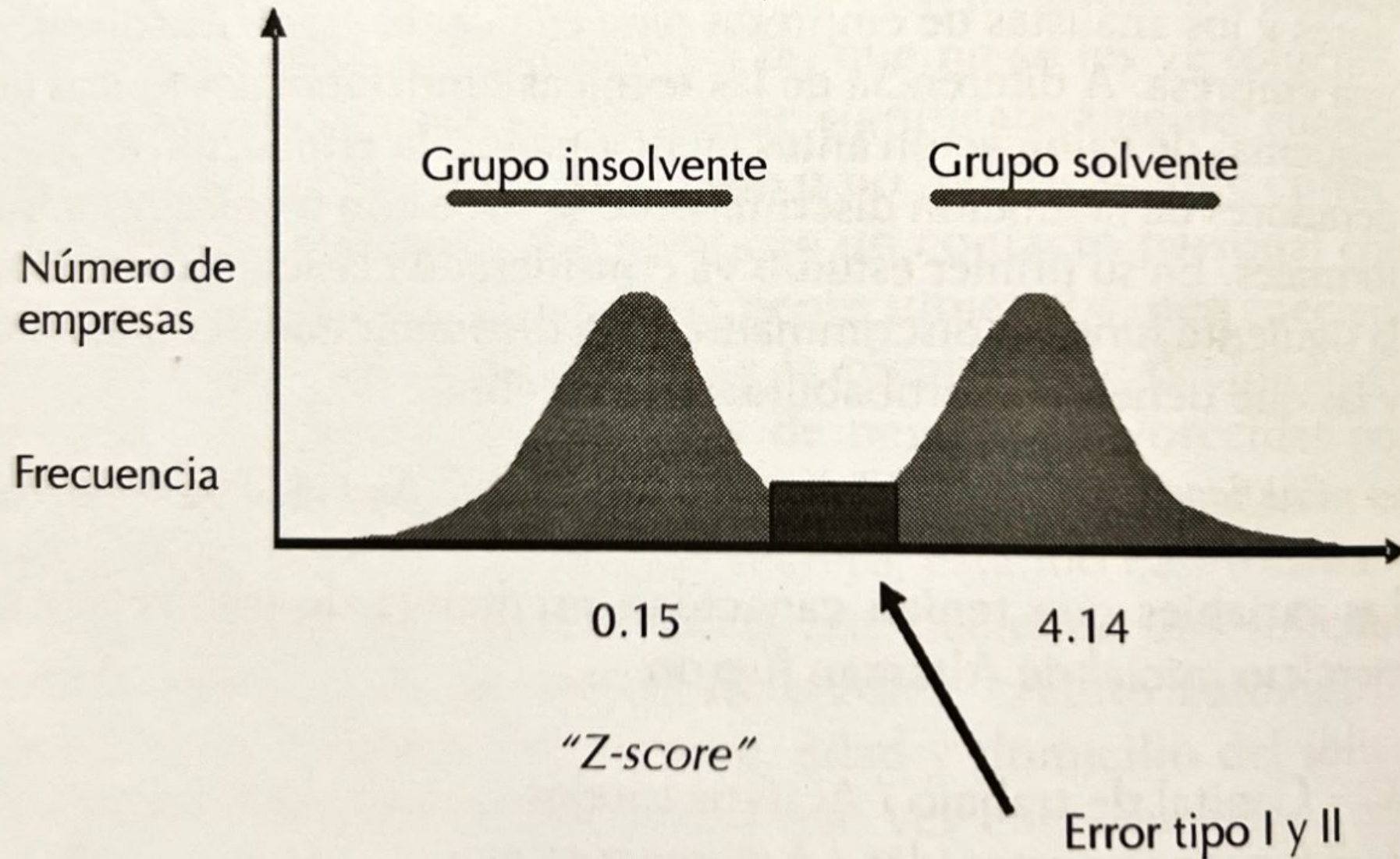
The Z-Score gained widespread popularity as a corporate bankruptcy predictor.

● **2000s**

Altman published revised models to improve the formula's accuracy.

FIGURA 3.3. CLASIFICACIÓN DE CRÉDITOS POR LA TÉCNICA "Z-SCORE" DE ALTMAN:
DISTRIBUCIONES DISCRIMINANTES Y LOS ERRORES DE TIPO I Y II.

FUENTE: ATMAN (1968)



Formula

It was developed in 1968 by Edward Altman and combines five financial ratios using a weighting system to calculate the overall Z-Score.

$$Z = 1.2x_1 + 1.4x_2 + 3.3x_3 + 0.6x_4 + 1.0x_5$$

Z-Score 2000

$$Z' = 0.717x_1 + 0.847X_2 + 3.107X_3 + 0.420X_4 + 0.998X_5$$

Z-Score Emerging Markets

$$Z'' = 6.56x_1 + 3.26x_2 + 6.72x_3 + 1.05x_4$$

A	B	
Particulars	Amount (in billion)	
ng Capital	\$14.4	
come	\$59.5	
nd Paid	\$13.7	
ed Earnings	\$45.8	
ting Income	\$70.9	
t Value of Equity	\$962.0	
Liabilities	\$258.6	
	\$265.6	
Assets	\$365.7	
, X ₃ , X ₄ , X ₅ is calculated as		
Notation	Value	For
X ₁	0.04	=B
X ₂	0.13	=B
X ₃	0.19	=B1
X ₄	3.72	=B1
X ₅	0.73	=B1

Variables

X1 Working Capital / Total Assets

X1 is the ratio of Working Capital divided by Total Assets. It measures a company's liquidity.

X2 Retained Earnings / Total Assets

X2 is the ratio of Retained Earnings divided by Total Assets. It measures profitability.

X3 EBIT / Total Assets

X3 is the ratio of Earnings Before Interest and Taxes (EBIT) divided by Total Assets. It measures operating efficiency.

X4 Market Value of Equity / Total Liabilities

X4 is the ratio of Market Value of Equity divided by Total Liabilities. It measures leverage.

X5 Sales / Total Assets

X5 is the ratio of Sales divided by Total Assets. It measures asset turnover.

Applications



Financial status

The Z-Score indicates financial solvency and predicts bankruptcy



Investment decisions

Investors use the Z-Score to evaluate investment risks



Lending decisions

Banks use the Z-Score to make lending decisions by assessing creditworthiness



Audits

Auditors use the Z-Score to evaluate business financial health

The Z-Score is a useful financial metric with many applications for assessing financial health.

Limitations

- **Does not consider qualitative factors**

The Altman Z-Score is solely based on financial ratios and does not take into account important qualitative factors like management experience, industry trends, competitive position, etc.

- **Backward-looking**

The ratios used in the model rely on historical financial data, so the score may not accurately predict sudden future changes in a company's performance or financial distress.

- **Industry biases**

The original model was developed using data from manufacturing companies, so it may not work as well for companies in other industries like retail, technology, etc.

- **Private companies**

The model was designed for public companies with available financial data, so it has limitations when applied to private companies with limited financial disclosures.



Notable Examples

Lehman Brothers was a global financial services firm founded in 1850 that went bankrupt during the financial crisis of 2008. Lehman had relied heavily on mortgage-backed securities and suffered losses as the subprime mortgage market collapsed in 2007-2008. Its failure was a seminal event in the global financial crisis.

	A	B	C
2			
3	Particulars	Amount (in billion)	
6	Working Capital	\$14.4	
7	Net Income	\$59.5	
8	Dividend Paid	\$13.7	
9	Retained Earnings	\$45.8	
10	Operating Income	\$70.9	
11	Market Value of Equity	\$962.0	
12	Total Liabilities	\$258.6	
13	Sales	\$265.6	
14	Total Assets	\$365.7	
15			
16	X_1, X_2, X_3, X_4, X_5 is calculated as		
17			
18	Notation	Value	Formula
19	X_1	0.04	=B6/B14
20	X_2	0.13	=B9/B14
21	X_3	0.19	=B10/B14
22	X_4	3.72	=B11/B12
23	X_5	0.73	=B13/B14
24			

Conclusion

The Altman Z-Score has proven useful for predicting bankruptcy risk, but it is not perfect. The model was developed decades ago and may not fully account for changes in accounting standards and business practices over time. Additionally, the Z-Score is based on financial ratios that can be manipulated by companies. So while a low Z-Score is cause for concern, it should be considered along with other qualitative factors when evaluating bankruptcy likelihood.